Task 1: Lab Setup & Environment Verification Report

Author	Vansh Singh

Course / Task	Foundation & Environment Setup (Task 1 / Days 1–12)
Environment	Oracle VirtualBox, Host-Only Network (192.168.56.0/24)
Attacker VM	Kali Linux (kali, 192.168.56.101)
Target VM	Metasploitable2 (192.168.56.102)

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1. Executive Summary

This report documents the design, deployment, and verification of an isolated penetration testing laboratory built inside Oracle VirtualBox. The lab consists of two virtual machines: a **Kali Linux** attacker workstation and an intentionally vulnerable **Metasploitable2** target. Both are attached to a **VirtualBox Host-Only network** (vboxnet0, subnet 192.168.56.0/24), which keeps all attacker-to-target traffic within the host machine and prevents exposure to external networks.

Verification covered four areas:

1. **IP configuration** on both VMs.
2. **Layer 3 connectivity** using ICMP.
3. **Traffic interception** using Wireshark on the attacker VM.
4. **Service discovery** using Nmap.

All checks completed successfully, confirming the environment is ready for the reconnaissance, scanning, and exploitation exercises in later tasks.

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2. Lab Architecture & Network Topology

2.1 Component Overview

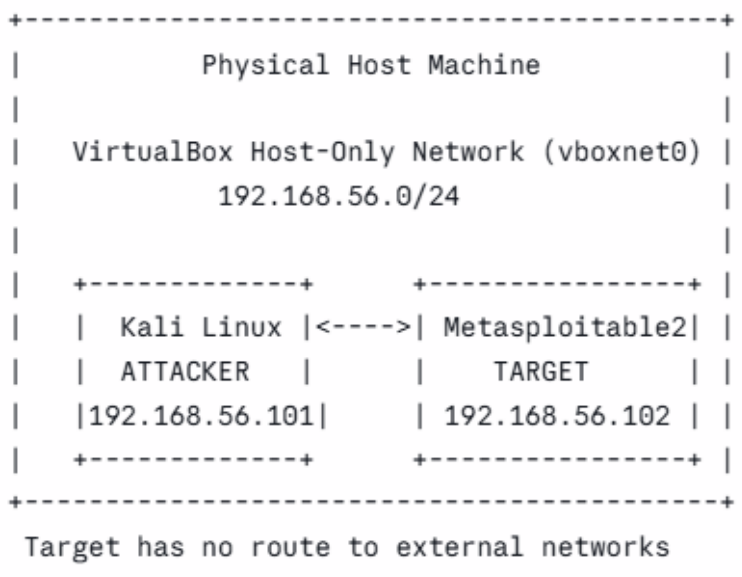
2.1 Component Overview

Component	Operating System	IP Address	Network Mode	Role
Kali Linux VM (`kali`)	Kali Linux (Debian-based)	`192.168.56.101` (`eth0`)	Host-Only Adapter (`vboxnet0`)	**Attacker**
Metasploitable2 VM	Ubuntu-based Linux (2.6.x kernel)	`192.168.56.102` (`eth0`)	Host-Only Adapter (`vboxnet0`)	**Target**
VirtualBox Host-Only Interface	Host OS virtual adapter (`VirtualBox Host-Only Ethernet Adapter`)	`192.168.56.1` (default for the segment)	Host-Only	Virtual switch / host interface

Note: The Kali VM also has a second adapter (eth1, 10.0.3.15/24) used for internet access (e.g., package updates). The Metasploitable2 VM has **only** the Host-Only adapter, so the target itself remains isolated. All attack and verification traffic in this report uses eth0 on 192.168.56.0/24.

2.2 Topology Diagram

text



2.3 Rationale for the Host-Only Adapter

Reason	Explanation
Isolation	Host-Only networking creates a private segment shared only between the host and the attached VMs. The vulnerable target cannot be reached from the LAN or the internet.
Prevention of external exposure	Metasploitable2 ships with deliberately insecure services (e.g., backdoored FTP, an open root bind shell, weak credentials). Keeping it off bridged or NAT networks avoids accidentally exposing it to real attackers.
Containment of attack traffic	Scans and exploit attempts generated from Kali stay inside the lab, avoiding unintended impact on third-party systems and keeping testing legal and ethical.
Clean packet capture	With only lab hosts on the segment, Wireshark captures contain just the relevant lab traffic, which makes protocol analysis simpler and more reliable.
Predictable addressing	A fixed subnet (192.168.56.0/24) makes scripting, documentation, and reproducibility straightforward.

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3. Step-by-Step Environment Verification

3.1 Verify Kali Linux Network Configuration

Objective: Confirm the Kali VM has an address in the Host-Only subnet.

┌──(kali㉿kali)-[~]

└─$ ip a

Observed output (relevant lines):

2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000

link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff

inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0

What was confirmed:

- eth0 state is UP.
- The inet address 192.168.56.101/24 belongs to the Host-Only subnet 192.168.56.0/24.
- The address is assigned dynamically (DHCP from the VirtualBox Host-Only DHCP server).
- A second interface, eth1 (10.0.3.15/24), is present for internet access and is not used for lab traffic.

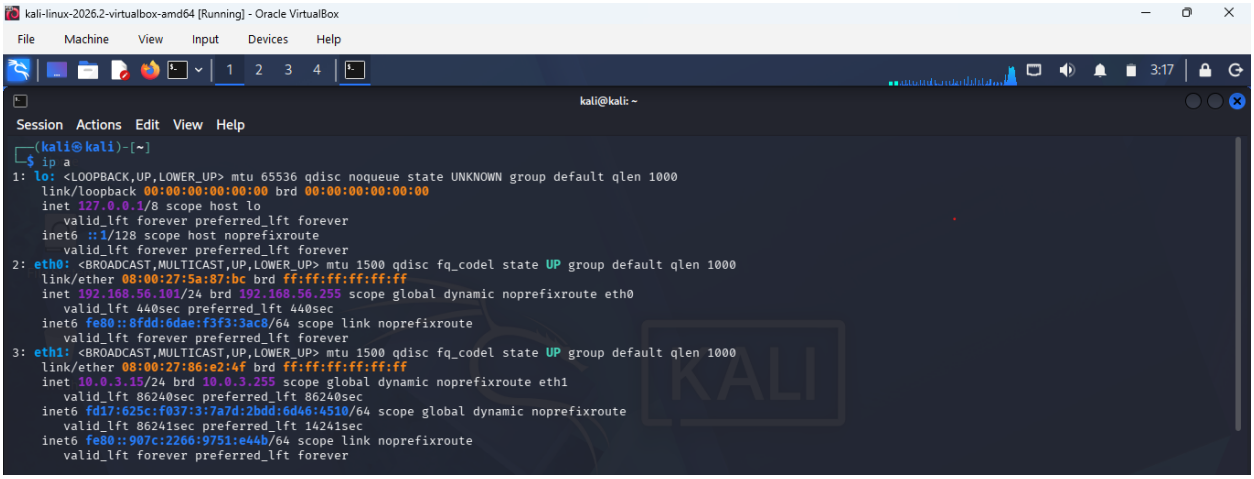


Figure 1: Kali Linux network interface configuration confirming Host-Only IP assignment.

3.2 Verify Metasploitable2 Network Configuration

Objective: Confirm the target VM is configured with its expected address.

msfadmin@metasploitable:~$ ifconfig

Observed output (relevant lines):

eth0 Link encap:Ethernet HWaddr 08:00:27:fd:61:07

inet addr:192.168.56.102 Bcast:192.168.56.255 Mask:255.255.255.0

UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1

What was confirmed:

- inet addr is 192.168.56.102.
- Mask is 255.255.255.0, matching the Kali VM's /24.
- Interface flags include UP and RUNNING.

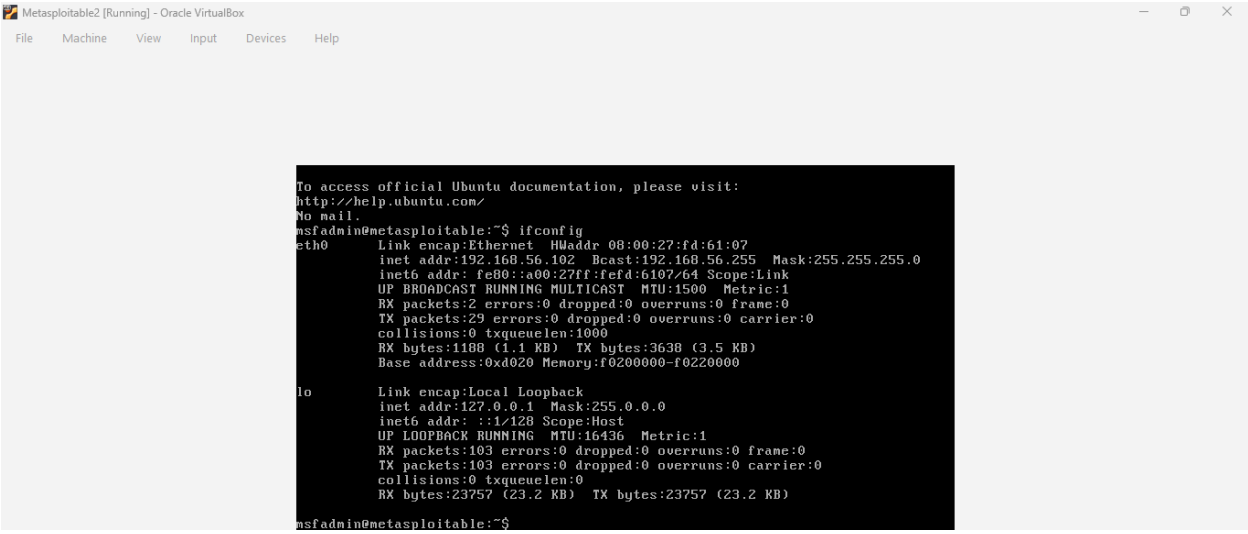


Figure 2: Metasploitable2 ifconfig output confirming the target IP address 192.168.56.102.

3.3 ICMP Connectivity Test (Kali → Metasploitable2)

Objective: Verify Layer 3 reachability from the attacker to the target.

┌──(kali㉿kali)-[~]

└─$ ping -c 4 192.168.56.102

Observed output:

PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.

64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=6.74 ms

64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=2.14 ms

64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=2.97 ms

64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=3.47 ms

--- 192.168.56.102 ping statistics ---

4 packets transmitted, 4 received, 0% packet loss, time 3006ms

rtt min/avg/max/mdev = 2.140/3.829/6.740/1.746 ms

What was confirmed:

- **0% packet loss** (4 transmitted, 4 received).
- ttl=64, consistent with a Linux target one hop away.
- Average round-trip time of about 3.8 ms, normal for a local virtual network (the first packet is slower due to ARP resolution).



Figure 3: ICMP connectivity test from Kali Linux to Metasploitable2 (192.168.56.102).

3.4 Wireshark Packet Capture of ICMP Traffic

Objective: Demonstrate that the attacker VM can intercept and inspect traffic on the lab segment.

Procedure:

1. Launch Wireshark on Kali (sudo wireshark or via the application menu).
2. Select the Host-Only interface (eth0) and start the capture.
3. Apply the display filter:
4. icmp
5. In a second terminal, run:
6. ping -c 4 192.168.56.102
7. Observe the Echo Request and Echo Reply pairs in the packet list.

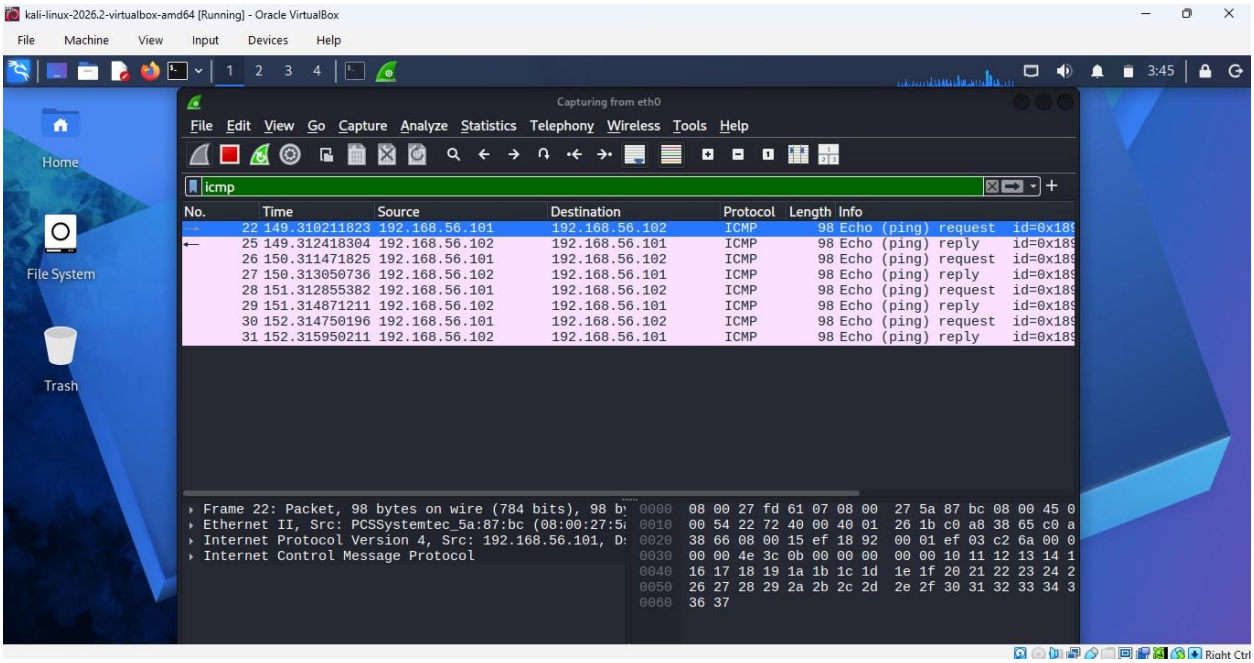


Figure 4: Wireshark ICMP capture showing request/reply streams between Kali VM (192.168.56.101) and Metasploitable2 (192.168.56.102).

3.5 Nmap Service Version Scan

Objective: Enumerate open ports and service versions on the target.

┌──(kali㉿kali)-[~]

└─$ nmap -sV 192.168.56.102

Observed output (abridged):

Starting Nmap 7.99 (https://nmap.org) at 2026-10-04 03:48 -0400

Nmap scan report for 192.168.56.102

Host is up (0.00083s latency).

Not shown: 977 closed tcp ports (reset)

PORT STATE SERVICE VERSION

21/tcp open ftp vsftpd 2.3.4

22/tcp open ssh OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)

23/tcp open telnet Linux telnetd

25/tcp open smtp Postfix smtpd

53/tcp open domain ISC BIND 9.4.2

80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)

111/tcp open rpcbind 2 (RPC #100000)

139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)

445/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)

512/tcp open exec netkit-rsh rexecd

513/tcp open login

514/tcp open shell Netkit rshd

1099/tcp open java-rmi GNU Classpath grmiregistry

1524/tcp open bindshell Metasploitable root shell

2049/tcp open nfs 2-4 (RPC #100003)

2121/tcp open ftp ProFTPD 1.3.1

3306/tcp open mysql MySQL 5.0.51a-3ubuntu5

5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7

5900/tcp open vnc VNC (protocol 3.3)

6000/tcp open X11 (access denied)

6667/tcp open irc UnrealIRCd

8009/tcp open ajp13 Apache Jserv (Protocol v1.3)

8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1

MAC Address: 08:00:27:FD:61:07 (Oracle VirtualBox virtual NIC)

Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Key services of interest:

Port	Service	Relevance
21/tcp	FTP (vsftpd 2.3.4)	Known backdoored version; target for later exploitation exercises
22/tcp	SSH (OpenSSH 4.7p1)	Remote administration; credential testing
80/tcp	HTTP (Apache 2.2.8)	Hosts intentionally vulnerable web applications
3306/tcp	MySQL 5.0.51a	Database service; weak/default credentials exercises
1524/tcp	Bindshell	Unauthenticated root shell by design

In total, **23 open TCP ports** were discovered, with 977 closed ports not shown.

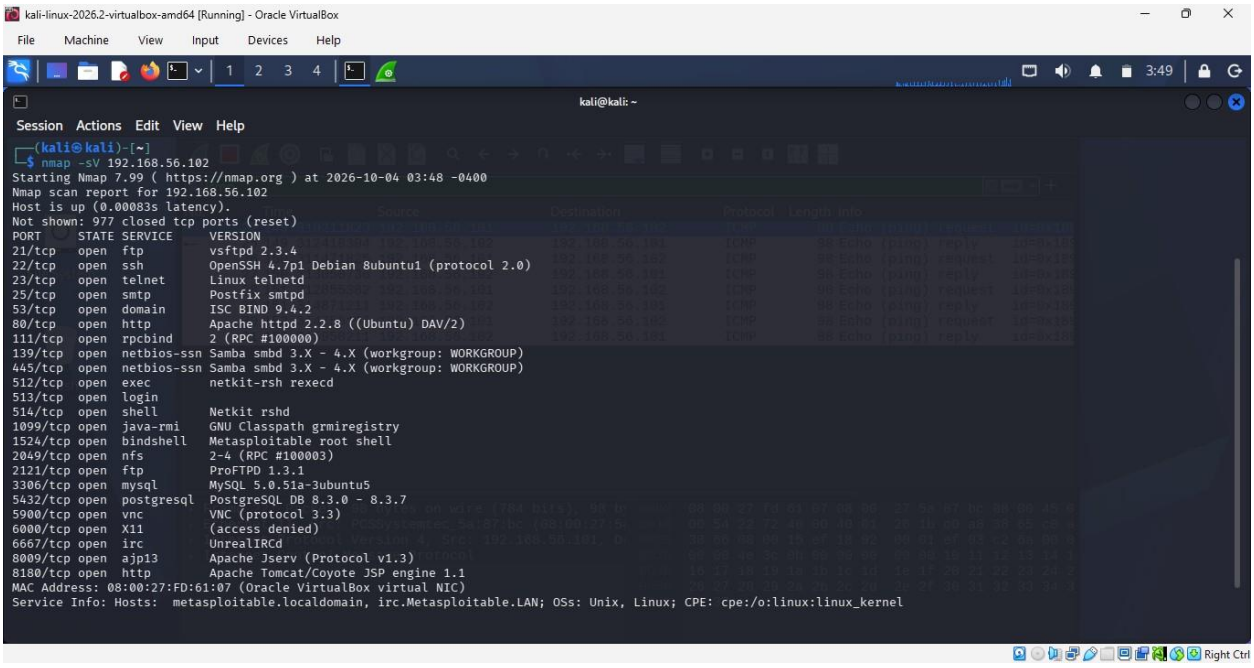


Figure 5: Nmap service version scan of Metasploitable2 revealing open ports and service banners.

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4. Wireshark Packet Capture Analysis

4.1 Packet Stream Breakdown

The capture was taken on eth0 with the display filter icmp. The table below summarizes the packets seen during the ping -c 4 test.

No.	Time (s)	Protocol	Source IP	Destination IP	Length	Frame / ICMP Type
22	149.310	ICMP	192.168.56.101 (Kali)	192.168.56.102	98	Echo (ping) **Request**, Type 8
25	149.312	ICMP	192.168.56.102	192.168.56.101 (Kali)	98	Echo (ping) **Reply**, Type 0
26	150.311	ICMP	192.168.56.101 (Kali)	192.168.56.102	98	Echo (ping) **Request**, Type 8
27	150.313	ICMP	192.168.56.102	192.168.56.101 (Kali)	98	Echo (ping) **Reply**, Type 0
28	151.312	ICMP	192.168.56.101 (Kali)	192.168.56.102	98	Echo (ping) **Request**, Type 8
29	151.314	ICMP	192.168.56.102	192.168.56.101 (Kali)	98	Echo (ping) **Reply**, Type 0
30	152.314	ICMP	192.168.56.101 (Kali)	192.168.56.102	98	Echo (ping) **Request**, Type 8
31	152.315	ICMP	192.168.56.102	192.168.56.101 (Kali)	98	Echo (ping) **Reply**, Type 0

Four request/reply pairs match the four packets sent by ping -c 4. The requests are spaced about one second apart, which is the default ping interval.

4.2 Frame Structure (Per Packet)

Expanding a packet in the details pane (for example, Frame 22: 98 bytes on wire, 784 bits) shows the following layers:

- **Frame:** Captured frame metadata (98 bytes, timestamp).
- **Ethernet II:** Source and destination MAC addresses (VirtualBox OUI 08:00:27). For Frame 22, the source is Kali (08:00:27:5a:87:bc) and the destination is Metasploitable2 (08:00:27:fd:61:07).
- **Internet Protocol Version 4:** Source 192.168.56.101, destination 192.168.56.102, protocol field = ICMP (1).

- **Internet Control Message Protocol:** Type, Code, Checksum, Identifier, Sequence Number, and Data payload.

4.3 Observations on ICMP Echo Request and Reply

Attribute	Echo Request	Echo Reply
ICMP Type	8	0
ICMP Code	0	0
Direction	Kali → Metasploitable2	Metasploitable2 → Kali
Source / Destination	192.168.56.101 → 192.168.56.102	192.168.56.102 → 192.168.56.101
Identifier / Sequence	Set by sender	Mirrors the matching request
Length	98 bytes	98 bytes

Key observations:

- Every Echo Request (Type 8) is followed by an Echo Reply (Type 0) with the **same identifier** (id=0x189…), confirming a clean bidirectional exchange.
- The reply is the same size as the request (98 bytes), as expected, since the target echoes the payload back unchanged.
- Request-to-reply delay is roughly 1–2 ms in the capture, consistent with the ping latency in Section 3.3.
- **No unexpected external IP addresses** appear in the filtered capture. All ICMP traffic is confined to 192.168.56.0/24.

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5. Security Findings & Conclusion

5.1 Verification Summary

Check	Method	Result
Attacker IP assignment	ip a on Kali	✅ 192.168.56.101/24 on eth0
Target IP assignment	ifconfig on Metasploitable2	✅ 192.168.56.102

Check	Method	Result
Bidirectional connectivity	ping -c 4 192.168.56.102	✅ 0% packet loss, avg RTT 3.83 ms
Packet interception	Wireshark icmp filter on eth0	✅ 4 Request/Reply pairs captured
Service discovery	nmap -sV 192.168.56.102	✅ 23 open ports, including FTP, SSH, HTTP, MySQL

5.2 Findings

- **Isolation confirmed for the target:** Metasploitable2 is attached only to the Host-Only network (192.168.56.0/24), so it is not reachable from the LAN or the internet.
- **Full bidirectional communication:** ICMP Echo Requests and Replies flow in both directions between the attacker and the target with no packet loss.
- **Interception capability validated:** Wireshark on the Kali VM captures and decodes lab traffic, confirming the environment supports later traffic analysis tasks.
- **Target exposure as expected:** Nmap reveals 23 open services on Metasploitable2, including legacy and intentionally vulnerable ones (vsftpd 2.3.4, an open root bindshell on port 1524, UnrealIRCd, Apache Tomcat, and outdated OpenSSH, Apache and MySQL versions). This is the intended attack surface for the exercises to follow.
- **Dual-homed attacker:** The Kali VM has an additional NAT-style adapter (eth1, 10.0.3.15) for internet access. This does not affect the target's isolation, but it should be kept in mind when interpreting any capture taken on eth1 rather than eth0.

5.3 Security Reminder

Metasploitable2 is **deliberately insecure** and must **never** be attached to a Bridged or NAT adapter, or otherwise exposed to untrusted networks. Keep the VM on the Host-Only adapter only.

5.4 Conclusion

The lab environment is correctly configured and fully operational. The attacker VM (Kali sLinux, 192.168.56.101) communicates bidirectionally with the target VM (Metasploitable2, 192.168.56.102), intercepts and analyzes packets with Wireshark, and enumerates services with Nmap. The environment satisfies the requirements of **Task 1** and is ready for the next phases: reconnaissance, vulnerability assessment, and exploitation.

Prepared by: Vansh Singh